Determining Critical Layers of Visual Models Using Logit Lens

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Abstract

While the logit lens technique has proven valuable for understanding layer-wise learning dynamics in language models, its application to computer vision remains underexplored. We extend the logit lens to vision transformers (ViTs) and propose a framework for identifying critical layers by combining logit lens analysis with three complementary metrics: cosine similarity, KL divergence and true class probability. We apply this approach to empirically validate theoretical predictions about layer importance while comparing ViTs trained using contrastive learning and self-supervised learning with knowledge distillation. Our analysis reveals a striking difference in semantic development: CLIP models develop class-relevant understanding significantly later than DINO models. By identifying layers that consistently emerge as critical across multiple metrics, we uncover fundamental differences in how contrastive multimodal versus self-supervised learning shape hierarchical representations. These findings provide empirical grounding for theoretical layer importance predictions and offer practical guidance for model optimization and architectural design.

Keywords: vision transformers, logit lens, critical layers, CLIP, DINO

1 Introduction

While the logit lens has proven effective for analyzing language models (nostalgebraist, 2022), the internal dynamics of vision transformers across different training paradigms are still being actively explored. Prior work suggests that CLIP cumulates its discriminative power in the final layers (Wang et al., 2024), while DINO models concentrate semantic features in the middle layers (Caron et al., 2021), but precise localization of critical processing has been lacking.

In this work, we address the following research questions: *Do vision transformer models exhibit distinct critical layers that significantly contribute to their final predictions, and how does the location of these layers differ between models trained with different paradigms (e.g., CLIP vs. DINO)? Additionally, how effective is the logit lens technique at identifying these critical layers?*

We extend the logit lens methodology to vision transformers and systematically analyze CLIP and DINO models to identify where task-relevant information emerges. Our findings highlight distinct patterns associated with different training methods and support the utility of the logit lens as a tool for interpreting vision transformers.

2 Related Work

The logit lens was introduced by nostalgebraist (2022) as a method for projecting intermediate transformer activations into output logit space, enabling analysis of layer-wise representation development in language models. This approach was refined by Belrose et al. (2023) through the Tuned Lens, which trains separate probes for each layer to improve projection fidelity. Recent work has extended these techniques to vision-language models: Jiang et al. (2024) applied logit lens methods to reduce hallucinations, while Joseph et al. (2025) incorporated them into interpretability toolkits.

Prior studies have provided theoretical insights into how training objectives affect ViT representations. Caron et al. (2021) demonstrated that DINO-trained models develop semantic features primarily in middle-to-late layers, while Wang et al. (2024) suggested that CLIP models localize discriminative power in final layers. However, these works lack precise quantification of where critical processing occurs.

Our work bridges this gap by applying the logit lens directly to vision transformers, enabling systematic comparison of critical layer locations across training paradigms and offering empirical support for theoretical claims about representation dynamics.

3 Methodology

To investigate the internal representations of CLIP and DINO models, we employ the logit lens technique.

3.1 Logit Lens for Vision Transformers

Given a visual transformer model f, we denote the output of the i-th transformer block as $h_i \in \mathbb{R}^d$, where d is the hidden dimension. For classification models like CLIP and DINO, the final output logits are obtained from the final block h_N followed by a projection head $W \in \mathbb{R}^{C \times d}$, where C is the number of classes. The logit prediction is as follows:

$$z_L = W h_L \tag{1}$$

In our adaptation of the logit lens framework, we repurpose the output projection head W to map an intermediate layer representation h_i directly into the logit space, producing:

$$z_i = W h_i \tag{2}$$

A schematic overview of the logit lens applied to a vision transformer is shown in Figure 1 in the appendix.

3.2 Comparative Metrics

To measure the similarity between the intermediate logits z_i and the final logits z_L, we use two standard measures: cosine similarity and Kullback–Leibler (KL) divergence.

Cosine Similarity This metric measures the angular similarity between the two logit vectors and is defined as:

$$\text{sim}(z_i, z_L) = \frac{z_i \cdot z_L}{\|z_i\|\|z_L\|} \tag{3}$$

It allows us to assess how similar the intermediate and final representations are in terms of their directional alignment.

Kullback–Leibler Divergence To assess the divergence in predicted distributions, we normalize logits using the softmax function and compute:

$$D_{\text{KL}}(p_L \parallel p_i) = \sum_{c=1}^{C} p_L(c) \log \left(\frac{p_L(c)}{p_i(c)} \right) \tag{4}$$

where $p_L = \text{softmax}(z_L)$ and $p_i = \text{softmax}(z_i)$. This measures how different the predicted distributions are.

3.3 Identifying Critical Layers

To identify *critical layers*, we use three metrics: cosine similarity, KL divergence, and true class probability. For each class, we compute a ranking of layers from most to least important based on how much each layer moves the prediction closer to the final output. Specifically, we look at the change in similarity to the final logits (cosine similarity), the reduction in distributional difference (KL divergence), and the increase in probability assigned to the true class (TCP). From each ranking, we select the top-3 and bottom-3 layers per class. The final importance scores are obtained by aggregating the frequency of each layer's appearance across all classes.

4 Experiments

We evaluate our method only on a stratified 10% subset of Tiny ImageNet (10,000 images of 200 classes) due to limitations in computational resources and processing power. The models tested are CLIP ViT-B/32, CLIP ViT-L/14 (both pretrained by OpenAI), and DINO ViT (from the `timm` library). For DINO, a linear classification head was trained on frozen features for 10 epochs using a different stratified 10% subset of Tiny ImageNet to enable logit extraction at intermediate layers. All models are evaluated in inference mode using the methods and metrics described in Section 3. Our experiments reveal distinct patterns in how vision transformers develop task-relevant representations across layers, with clear differences between training paradigms.

4.1 Layer-wise Representation Evolution

Figure 2 in the appendix reveals processing locations characteristic of each training approach. CLIP models exhibit gradual progression with dramatic final-layer jumps: ViT-B/32 shows sharp cosine similarity increases in the final layer, while ViT-L/14 distributes this across layers 20-24. Both demonstrate similar patterns—KL divergence drops significantly in final layers, and true class probabilities rise only in the last 2-3 layers.

DINO concentrates critical processing earlier: cosine similarity starts lower (0.05) and rises drastically around layer 9, then continues growing at a much slower rate through

final layers. True class probabilities begin rising at layers 7-8, indicating discriminative processing occurs in middle-to-late layers (7-10), not at the network's end like CLIP.

4.2 Critical Layer Identification

Figure 3 in the appendix shows that CLIP model's criticality lays in final layers (10-11), reflecting contrastive learning where final layers perform multimodal alignment. DINO exhibits concentrated criticality around layer 9, suggesting self-supervised learning consolidates semantic information at intermediate depths. The logit lens technique shows moderate effectiveness, with cosine similarity providing consistent results. Increased disagreement appears primarily with true class probability metrics across models and KL-divergence specifically in DINO, revealing underlying representational complexity.

To take the analysis one step further and provide both a deeper and holistic view of layer behavior under each metric, we scored each layer based on its frequency and rank among the top-3 most or least critical layers across models and metrics, while also examining the average change in each layer. The results are shown in Table 1 of the appendix.

5 Conclusion

We found that vision transformers do have critical layers, but their locations depend on the training process. In CLIP models, critical layers appear at the end, reflecting alignment-focused contrastive learning. In DINO, they occur earlier, closer to the middle, possibly due to semantic consolidation from self-supervised learning.

The logit lens proves effective—especially when paired with cosine similarity—but shows limitations in distinguishing clear ciritical layers when paired with metrics like KL divergence and true class probability, particularly in more complex representations.

These findings suggest that model optimization should focus on paradigm-specific layers: late layers in CLIP, and middle layers in DINO. Future work could expand to other models, datasets, and improved probing methods.

6 Reproducibility Statement

To facilitate reproducibility and transparency, we have made all code and relevant documentation publicly available on GitHub at `https://github.com/kaminskaz/wb-projekt`.

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7 Appendix

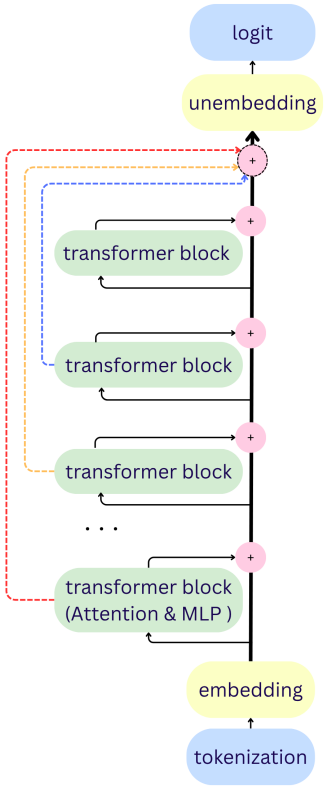


Figure 1: Schema of the Logit Lens technique applied to a vision transformer.

Model	Layer	Cosine	Cosine (avg ± std)	KL	KL (avg ± std)	TCP	TCP (avg ± std)
CLIP (ViT-B/32)	1	-452	-0.003 ± 0.010	-136	0.005 ± 0.062	-166	0.000 ± 0.000
	2	-	0.022 ± 0.005	17	-0.118 ± 0.052	-176	0.000 ± 0.000
	3	-115	0.014 ± 0.006	-121	-0.005 ± 0.044	-88	0.000 ± 0.000
	4	11	0.031 ± 0.007	33	-0.138 ± 0.056	-52	0.000 ± 0.001
	5	63	0.034 ± 0.009	-24	-0.046 ± 0.065	-124	0.000 ± 0.001
	6	175	0.039 ± 0.009	129	-0.180 ± 0.103	-46	0.000 ± 0.001
	7	-86	0.015 ± 0.008	-333	0.042 ± 0.098	-172	0.001 ± 0.001
	8	-543	-0.010 ± 0.011	13	-0.116 ± 0.090	-121	0.000 ± 0.001
	9	**204**	0.040 ± 0.008	-567	0.142 ± 0.102	4	0.002 ± 0.002
	10	143	0.035 ± 0.018	**389**	-0.458 ± 0.199	**385**	0.020 ± 0.021
	11	**600**	0.304 ± 0.024	**600**	-2.832 ± 0.372	**556**	0.389 ± 0.287
CLIP_L (ViT-L/14)	1	-5	0.011 ± 0.004	-7	0.132 ± 0.067	-21	-0.000 ± 0.000
	2	-1	0.014 ± 0.005	-30	0.193 ± 0.071	-26	0.000 ± 0.000
	3	-368	0.000 ± 0.004	6	-0.286 ± 0.054	-4	0.000 ± 0.000
	4	-152	0.005 ± 0.005	-1	0.002 ± 0.068	-20	0.000 ± 0.000

	5	-	0.015 ± 0.006	-18	0.185 ± 0.084	-43	0.000 ± 0.001
	6	-119	0.006 ± 0.006	60	-0.393 ± 0.081	-6	0.000 ± 0.001
	7	-3	0.014 ± 0.006	-7	0.122 ± 0.079	-47	0.000 ± 0.001
	8	-	0.016 ± 0.008	-	-0.070 ± 0.108	-29	-0.000 ± 0.002
	9	-	0.023 ± 0.009	22	-0.330 ± 0.132	-29	0.000 ± 0.002
	10	-423	-0.002 ± 0.011	-399	0.489 ± 0.206	-123	-0.000 ± 0.003
	11	-3	0.018 ± 0.012	-32	0.152 ± 0.246	-86	0.000 ± 0.004
	12	-21	0.015 ± 0.012	-444	0.544 ± 0.363	-103	-0.000 ± 0.004
	13	-102	0.008 ± 0.012	**389**	-0.755 ± 0.284	-22	0.000 ± 0.004
	14	-1	0.020 ± 0.011	8	-0.262 ± 0.179	-114	0.000 ± 0.003
	15	1	0.032 ± 0.010	-2	0.042 ± 0.184	-99	0.001 ± 0.004
	16	-2	0.019 ± 0.011	-66	0.220 ± 0.224	-83	0.002 ± 0.005
	17	170	0.055 ± 0.011	103	-0.440 ± 0.219	-12	0.004 ± 0.008
	18	-	0.022 ± 0.010	-	-0.068 ± 0.195	-86	0.006 ± 0.012
	19	1	0.032 ± 0.011	-2	-0.050 ± 0.253	57	0.031 ± 0.040
	20	-	0.023 ± 0.012	-6	0.002 ± 0.289	43	0.041 ± 0.070
	21	**294**	0.059 ± 0.012	-1	-0.175 ± 0.370	160	0.066 ± 0.102
	22	134	0.054 ± 0.015	-173	0.286 ± 0.698	**290**	0.148 ± 0.183
	23	**600**	0.122 ± 0.017	**600**	-3.499 ± 0.843	**403**	0.259 ± 0.236
DINO	1	-561	-0.015 ± 0.027	-26	-0.590 ± 0.805	-224	0.000 ± 0.003
	2	-435	0.000 ± 0.016	-194	-0.272 ± 0.629	-236	0.000 ± 0.002
	3	-136	0.041 ± 0.016	-309	-0.018 ± 0.483	-229	0.000 ± 0.001
	4	-53	0.047 ± 0.011	-84	-0.431 ± 0.439	-100	0.000 ± 0.001
	5	-1	0.062 ± 0.012	-198	-0.190 ± 0.432	-169	0.001 ± 0.002
	6	-1	0.063 ± 0.013	-92	-0.406 ± 0.424	-91	0.002 ± 0.004
	7	14	0.093 ± 0.014	31	-0.768 ± 0.538	-25	0.010 ± 0.012
	8	**371**	0.154 ± 0.019	**247**	-1.289 ± 0.772	70	0.038 ± 0.051
	9	**600**	0.324 ± 0.026	**501**	-2.209 ± 1.300	**419**	0.225 ± 0.259
	10	**215**	0.123 ± 0.023	**210**	-1.224 ± 1.105	**453**	0.232 ± 0.271
	11	-13	0.057 ± 0.016	-86	-0.304 ± 0.418	132	0.081 ± 0.178

Table 1: This table summarizes layer importance scores and average change metrics across models and evaluation methods. For each model and metric (Cosine, KL, TCP), layers were assigned importance scores based on their presence in the top-3 most or least critical layers: being the most critical layer gives +3, second +2, third +1; similarly, being the least critical gives -3, -2, and -1 respectively. Layers not appearing in any top or bottom 3 list are marked with a "–". Additionally, the table includes the average (avg) and standard deviation (std) of the change rate for each layer, computed over all samples, not just those that appeared in critical rankings. This provides both a focused and holistic view of layer behavior under each metric. Bold scores indicate importance values greater than a threshold of 200, chosen because it reflects consistent identification of a layer among the top-3 across evaluation cases. Highlighted layers are considered the most consistently important, as all three metrics identified them as significant.

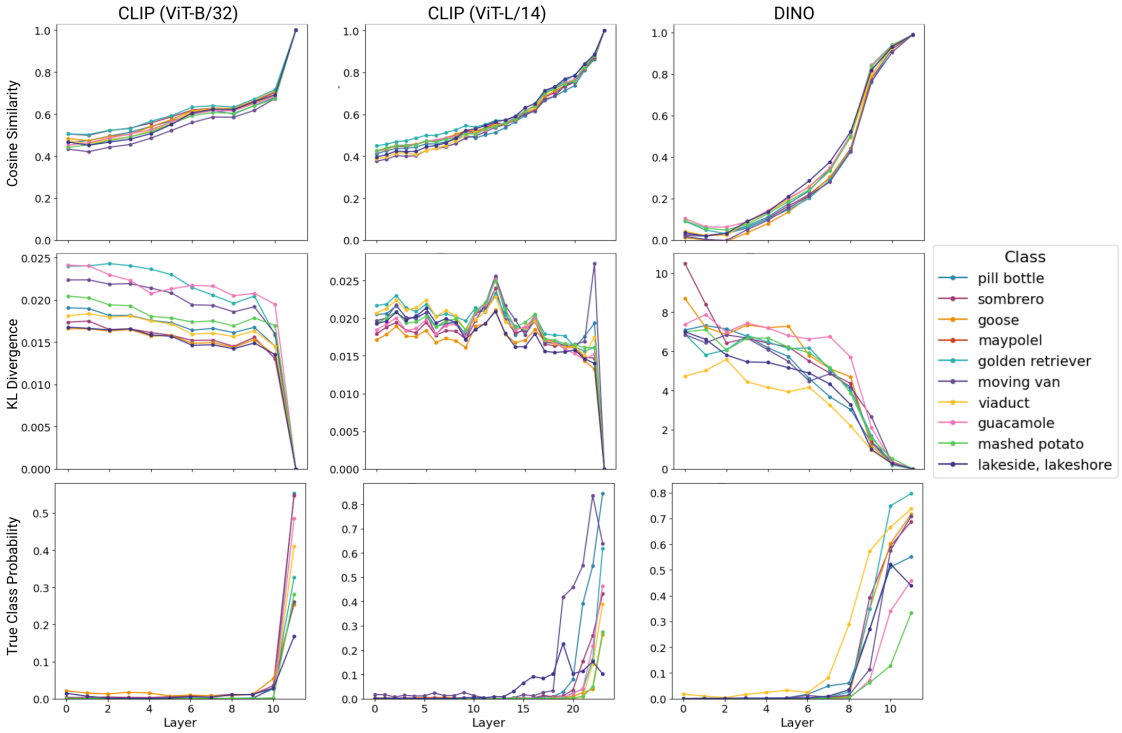


Figure 2: Metric trajectories across transformer layers for 10 selected classes (averaged per class) for . Rows show: Cosine Similarity (top), KL Divergence (middle), and True Class Probability (bottom). Columns correspond to different models: CLIP ViT-B/32, CLIP ViT-L/14, and DINO.

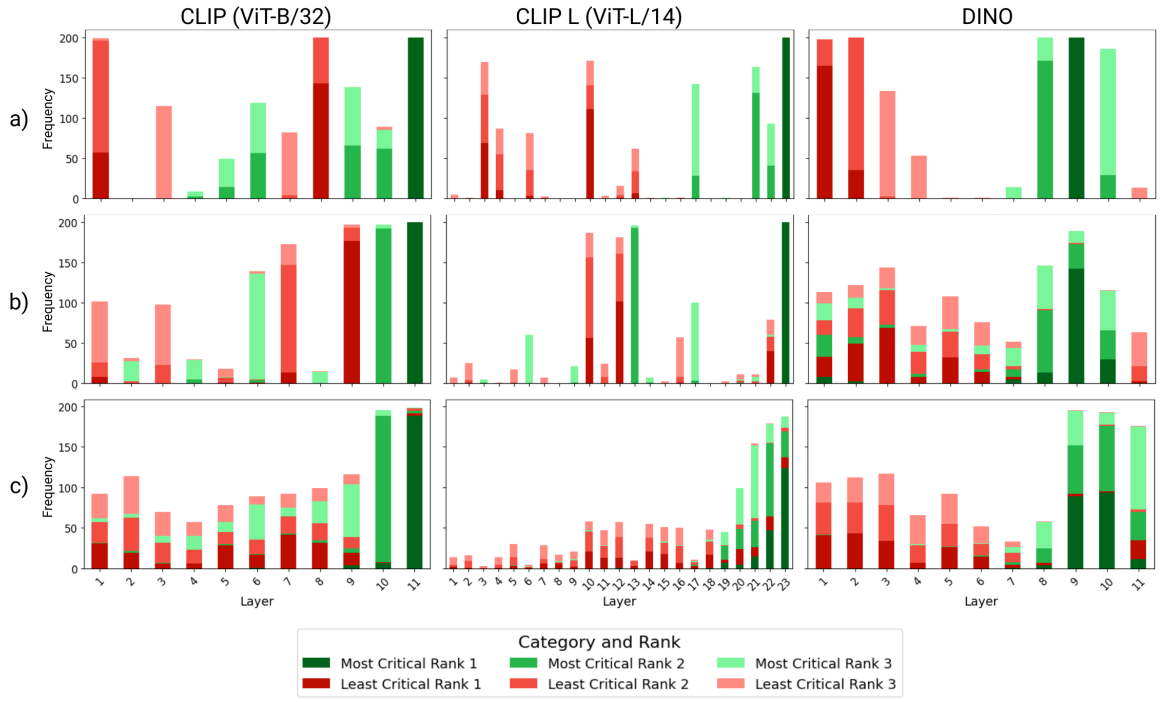


Figure 3: Distribution of most and least critical layers across all classes, estimated using (a) Cosine Similarity, (b) KL Divergence, and (c) True Class Probability. Stacked bars indicate how frequently each layer was selected as one of the top-3 or bottom-3 critical layers across classes. Columns correspond to different models: CLIP ViT-B/32, CLIP ViT-L/14, and DINO.